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Jahangiri et al.

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(54) **AUDIENCE FILTERING SYSTEM**

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(57) **ABSTRACT**

The present invention relates to improvements to systems and methods for filtering a data set. An audience filtering system is configured to perform operations that include accessing a database that comprises user profile data associated with a plurality of users, generating a bit-set based on the user profile data from the database, wherein the bit-set comprises an array of binary values, receiving a user input e.g., via a GUI) that defines one or more filter criteria (e.g., a value of a user attribute), filtering the bit-set based on the one or more filter criteria, and identifying a subset of the plurality of users based on the filtered bit-set, according to certain example embodiments.

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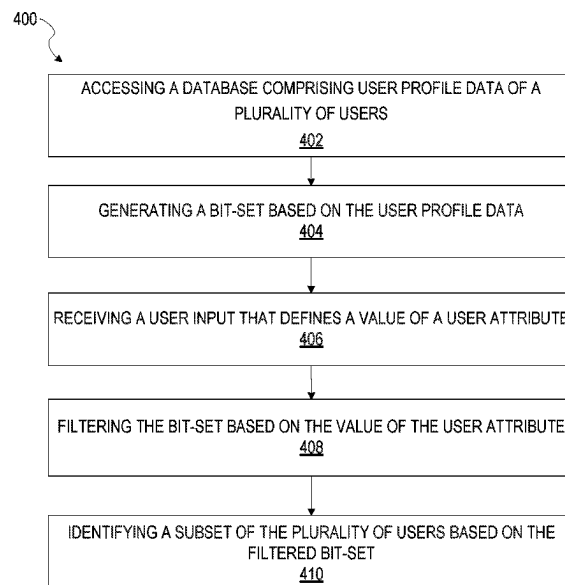
CPC **G06F 16/9535** (2019.01); **G06F 3/0482** (2013.01); **G06F 3/04847** (2013.01); **G06F 16/248** (2019.01); **G06F 16/29** (2019.01)

(58) **Field of Classification Search**

CPC G06F 16/9535; G06F 16/248; G06F 16/29; G06F 3/0482; G06F 3/04847

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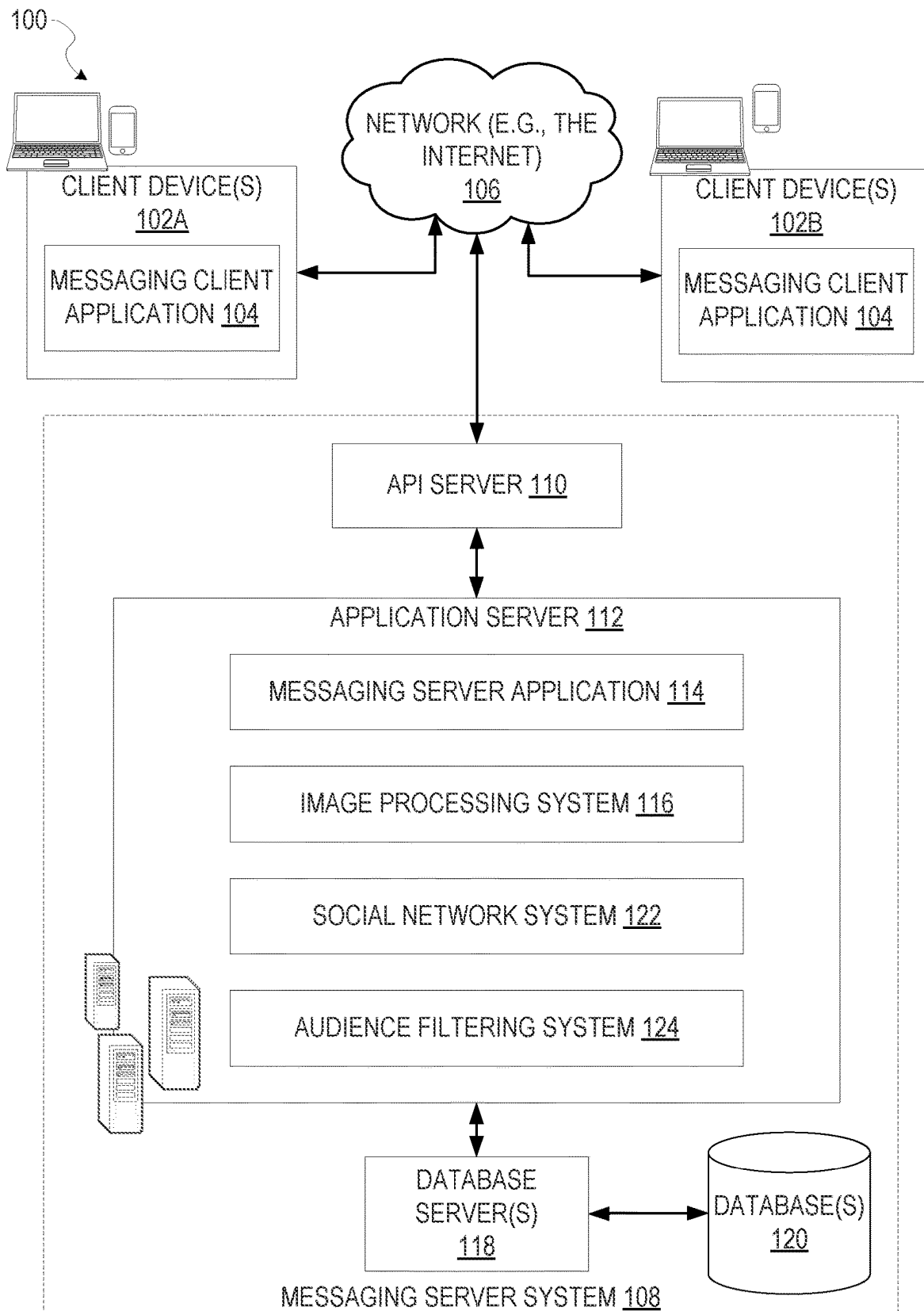


FIG. 1

100

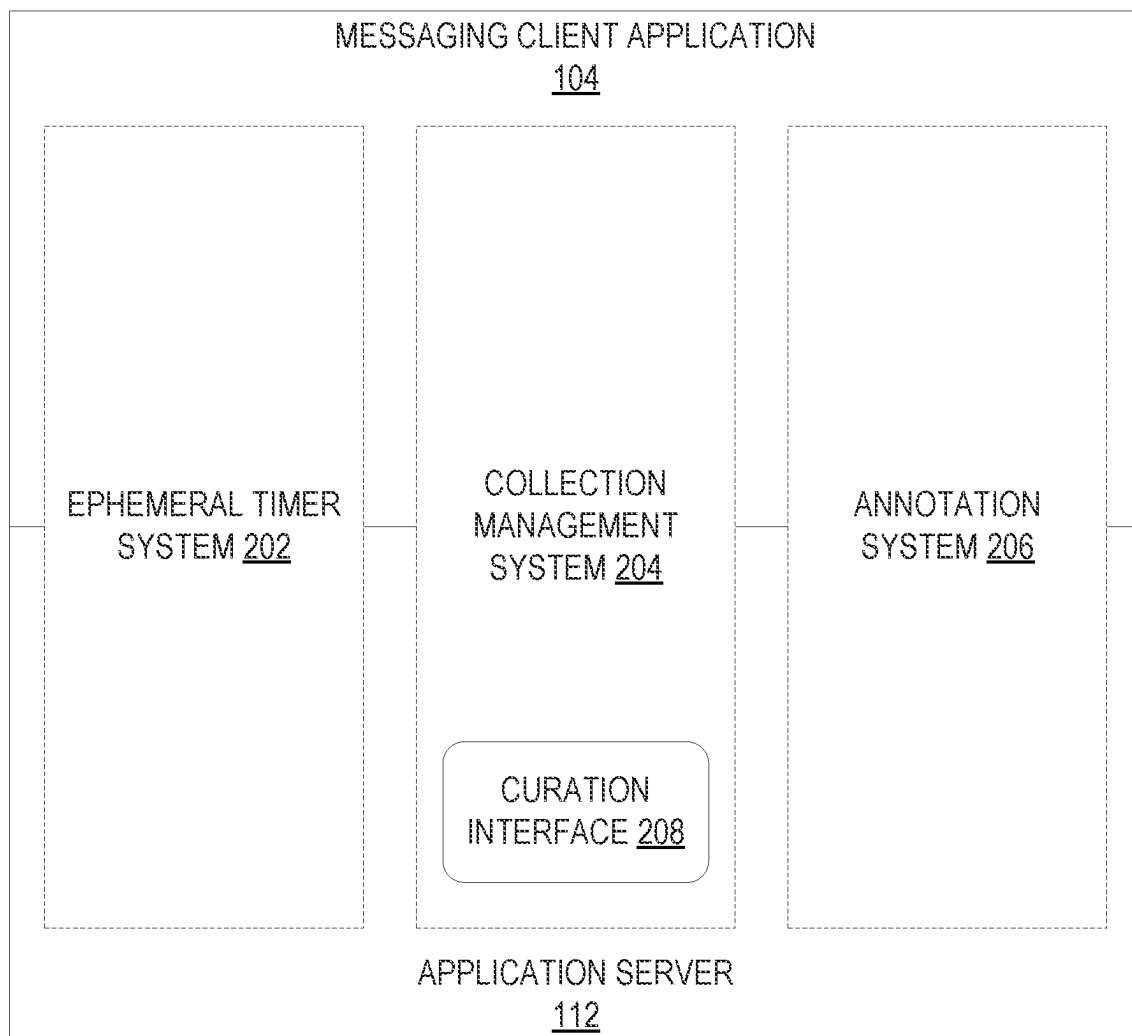



FIG. 2

300

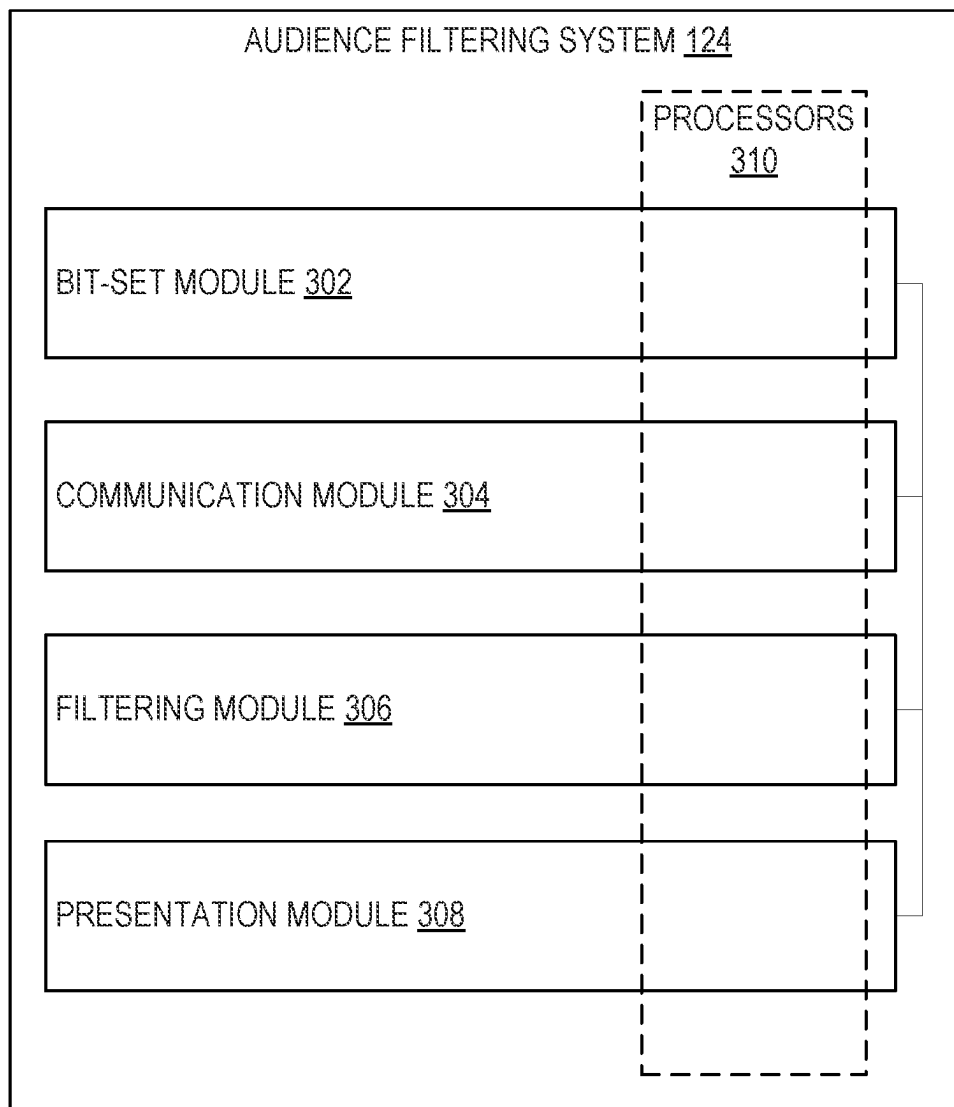



FIG. 3

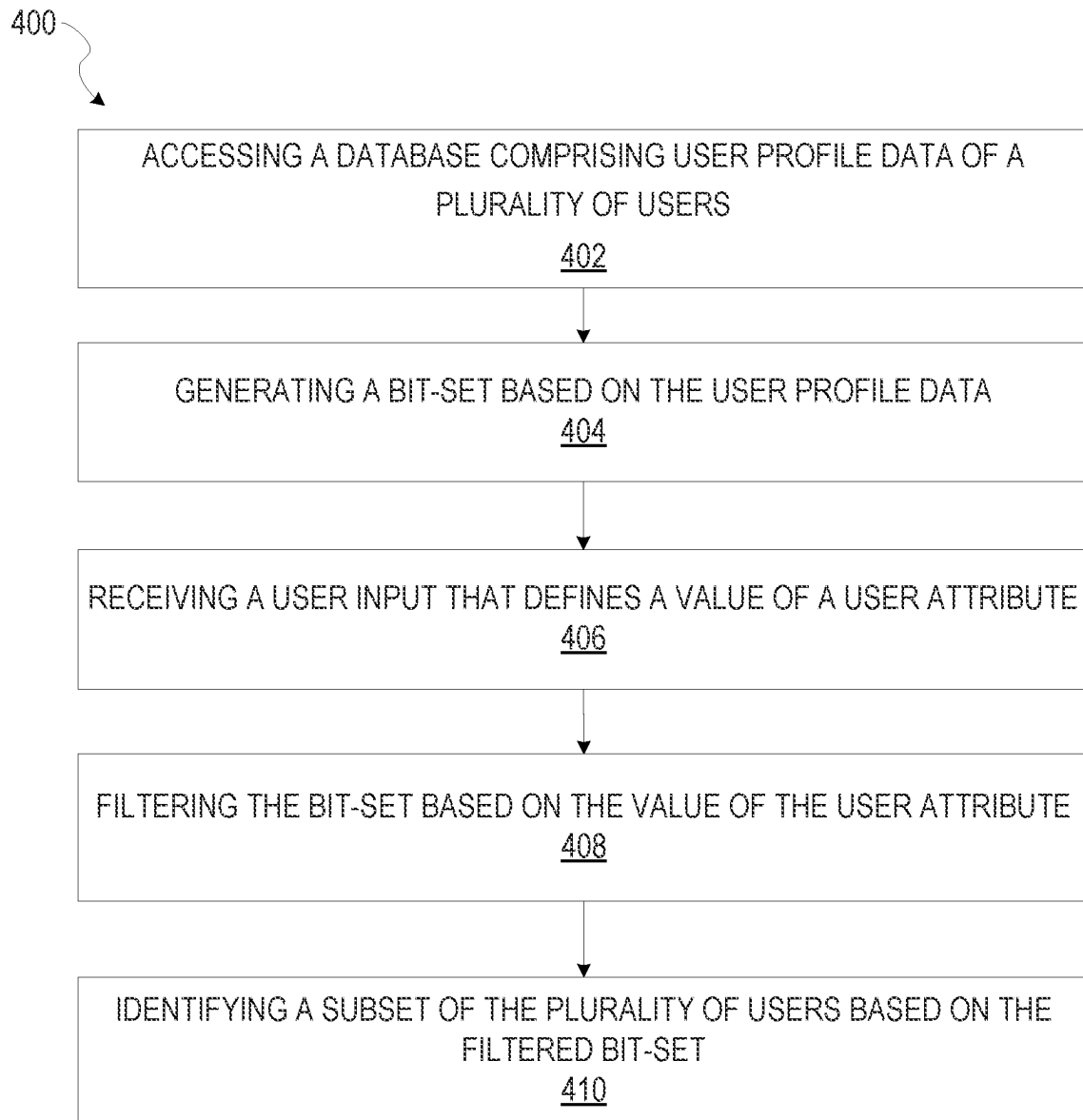
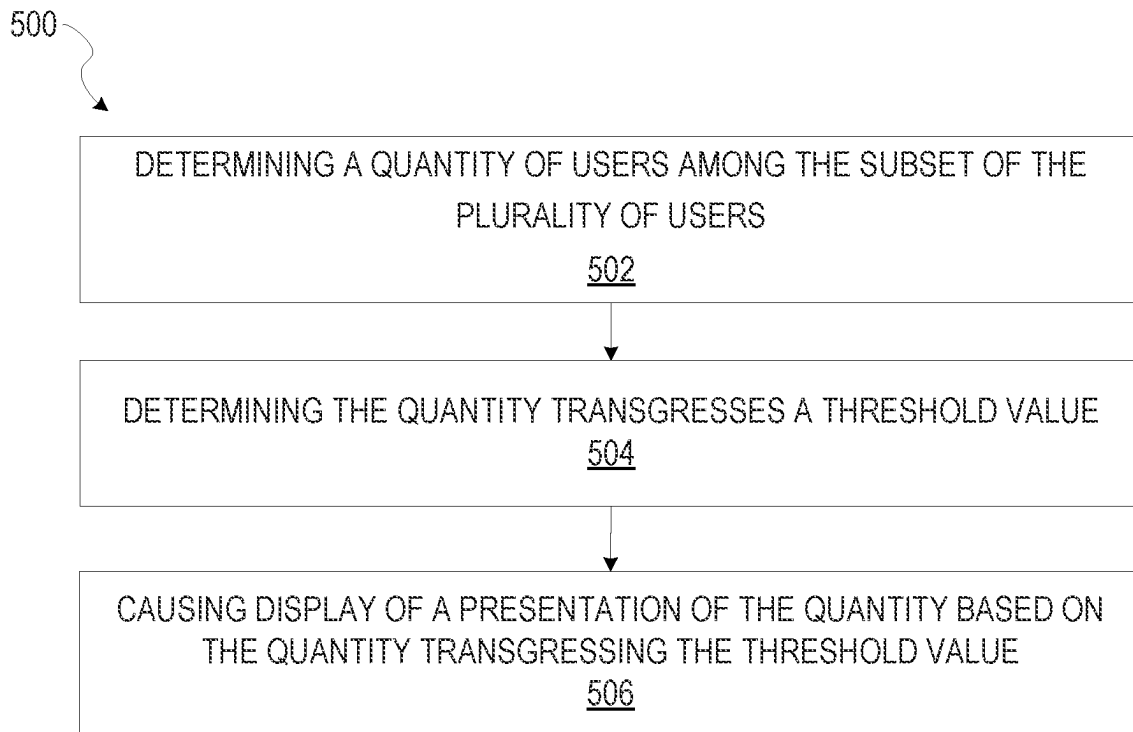
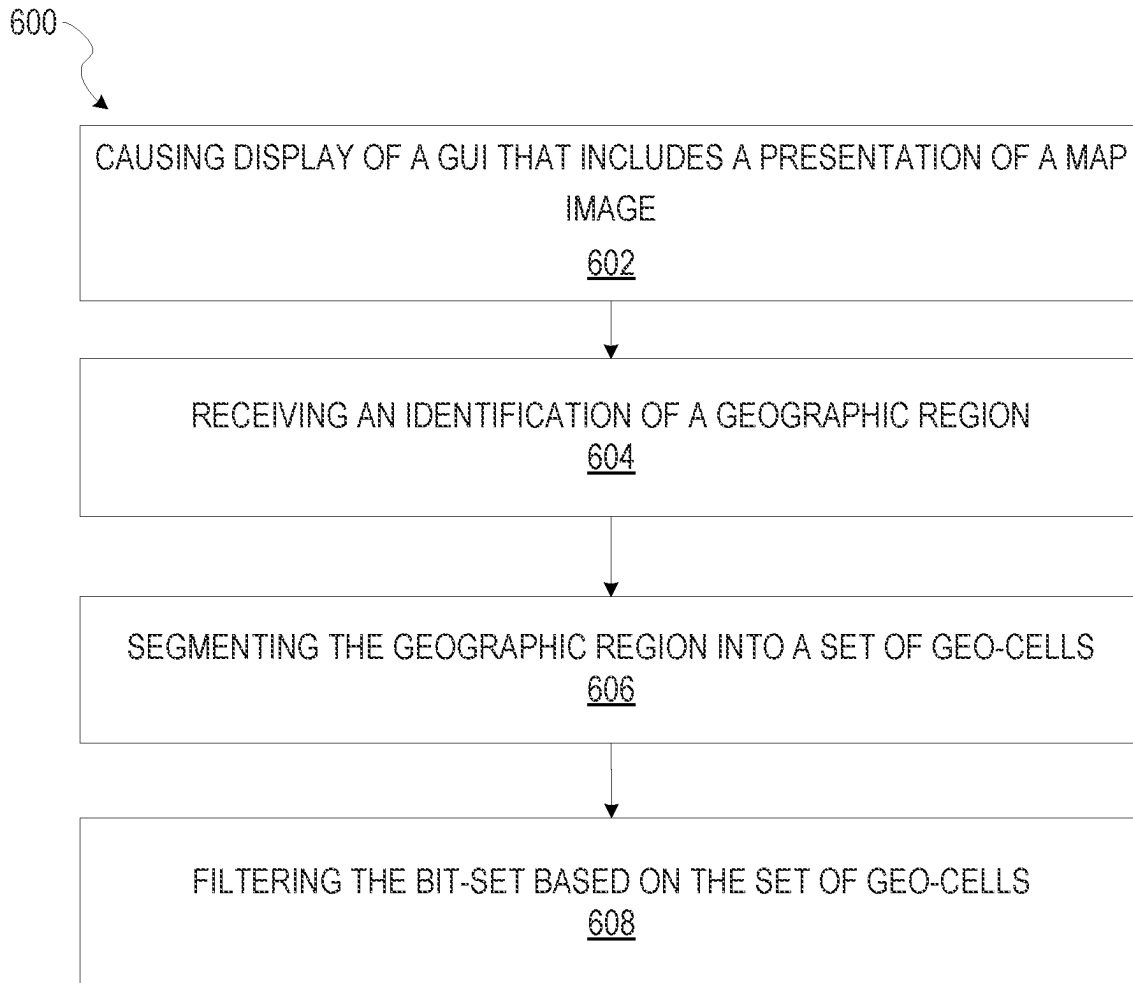


FIG. 4

*FIG. 5*

*FIG. 6*

700

	USER A				USER B			
GENDER: FEMALE	1	1	1	1	1	1	1	1
GENDER: MALE	1	1		1	1	1	1	1
AGE_GROUPS: 13-17								
...								
AGE_GROUPS: 35+	1	1	1	1	1	1	1	1
...								
AGE_GROUPS: 35+	1	1		1	1	1	1	1
...								
DEVICE: APPLE E-PHONE 7S	1	1	1	1	1	1	1	1
...								
LOCATION: SAN JOSE	1	1	1	1	1	1	1	1

FIG. 7

800

AD MANAGER BUSINESS MANAGER

CAMPAIGN
DRIVE VIDEO VIEWS

AD SET
AUDIENCE, SCHEDULE
& BUDGET

ADS
CREATIVES

CHOOSE AN AUDIENCE, BUDGET AND SCHEDULE

CREATE A SINGLE AD SET

CREATE A MULTIPLE AD SETS

GEOGRAPHY

LOCATIONS UNITED STATES

INCLUDE

REGION

SELECT LOCATION...

DEMOGRAPHICS

AGES

ALL 13-17 18-20 21-24 25-34 35+

GENDERS

ALL MALE FEMALE

LANGUAGES

SELECT CATEGORY...

DLX ADVANCED DEMOGRAPHICS

SELECT CATEGORY...

REGULATED CONTENT

☐ AGE RESTRICTIONS REQUIRED

AUDIENCES

PREDEFINED AUDIENCES

SELECT CATEGORY

CUSTOM AUDIENCES

INCLUDE

SELECT AUDIENCE...

PLACEMENTS

DELIVERY LOCATION

☒ ALL SNAPCHAT

☐ CONTENT PLACEMENT

DEVICES

OPERATING SYSTEMS

ALL ANDROID IOS

DEVICE MAKES

SELECT DEVICE MAKES...

AUDIENCE SIZE

28,750,000 - 33,350,000 PEOPLE

CANCEL

NEXT

802

802

804

FIG. 8

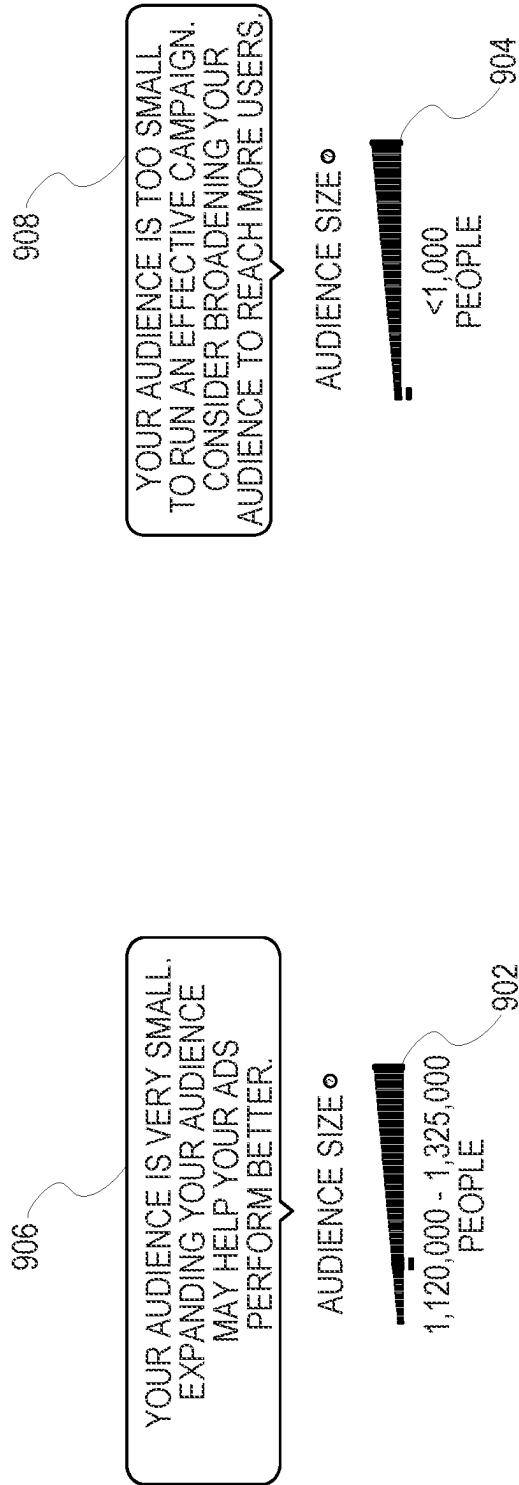


FIG. 9

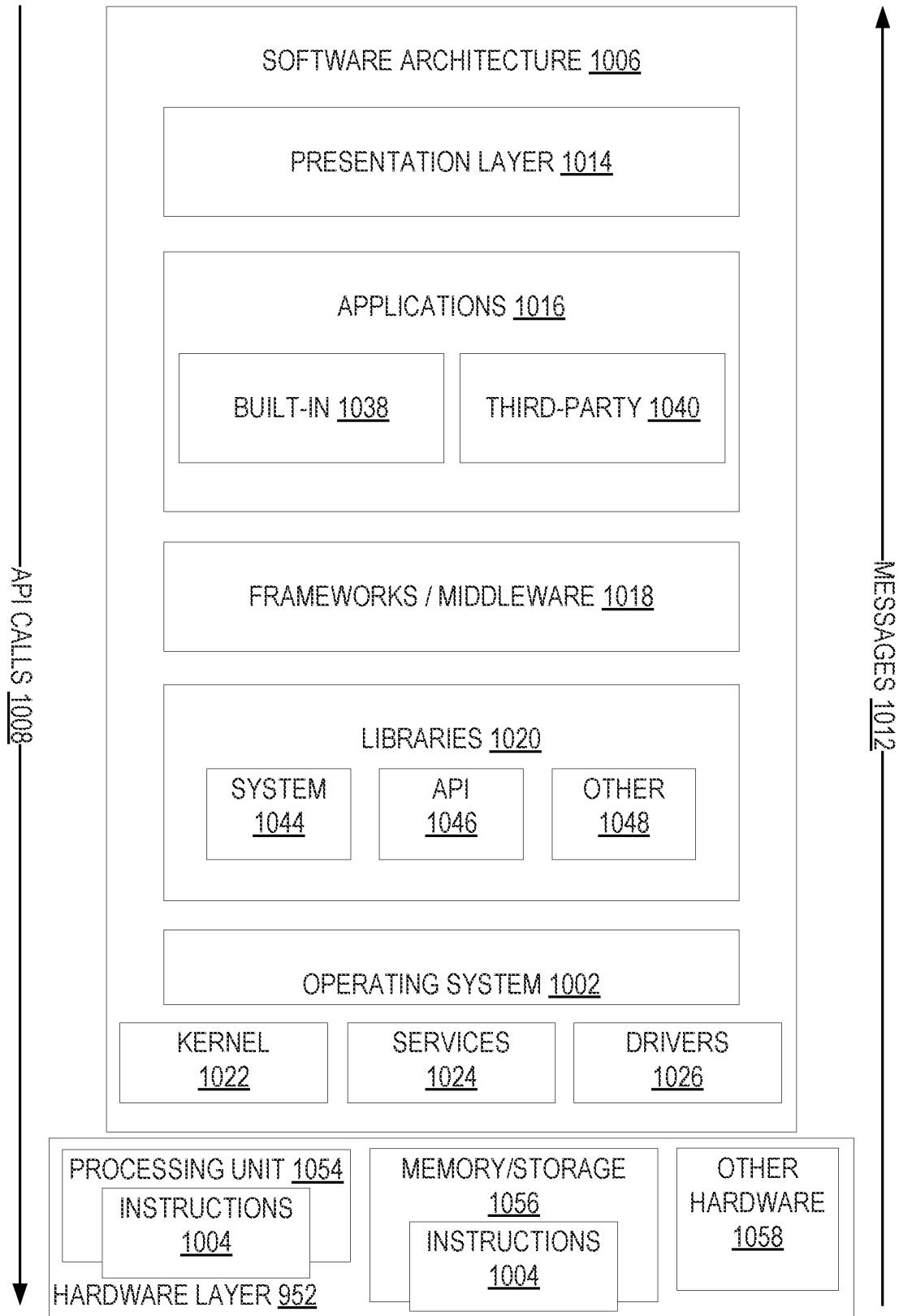


FIG. 10

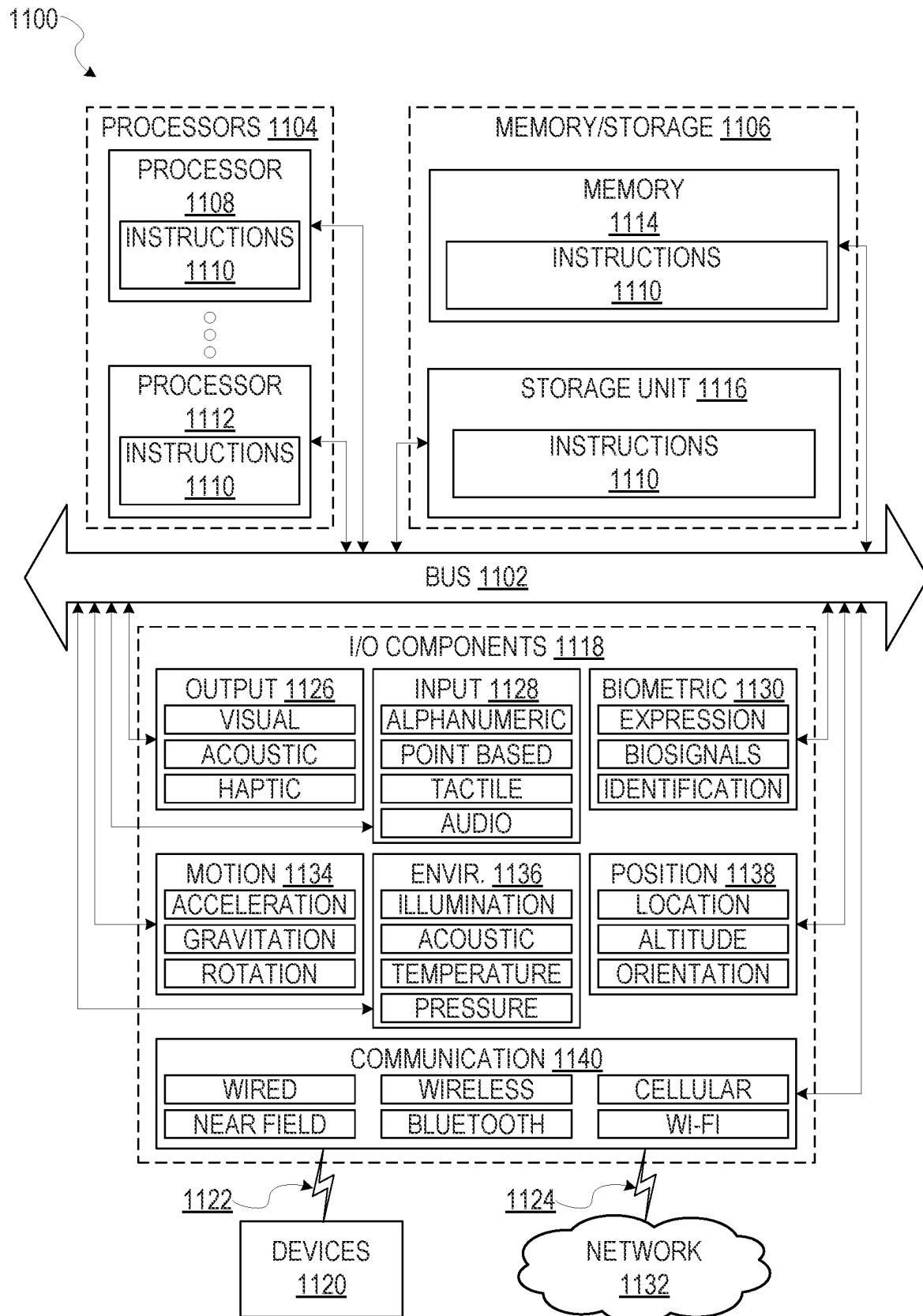


FIG. 11

1

AUDIENCE FILTERING SYSTEM**CLAIM OF PRIORITY**

This application is a continuation of U.S. patent application Ser. No. 15/908,559, filed on Feb. 28, 2018, which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to computing technology and, more particularly, but not by way of limitation, to systems for filtering datasets.

BACKGROUND

A bit-set, also known as a bit array, is an array data structure that compactly stores bits and can be used to implement a simple set data structure. A bit-set is a mapping from some domain (almost always a range of integers) to values in the set $\{0, 1\}$. The values can be interpreted as dark/light, absent/present, locked/unlocked, valid/invalid, etcetera. Thus, because there are only two possible values, so they can be stored in one bit. As with other arrays, the access to a single bit can be managed by applying an index to the array. Assuming its size (or length) to be n bits, the array can be used to specify a subset of the domain (e.g. $\{0, 1, 2, \dots, n-1\}$), where a 1-bit indicates the presence and a 0-bit the absence of a number in the set. This set data structure uses about n/w words of space, where w is the number of bits in each machine word.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

FIG. 1 is a block diagram showing an example messaging system for exchanging data (e.g., messages and associated content) over a network in accordance with some embodiments, wherein the messaging system includes an audience filtering system.

FIG. 2 is block diagram illustrating further details regarding a messaging system, according to example embodiments.

FIG. 3 is a block diagram illustrating various modules of an audience filtering system, according to certain example embodiments.

FIG. 4 is a flowchart illustrating a method for filtering data, according to certain example embodiments.

FIG. 5 is a flowchart illustrating a method for causing display of an indication of a quantity of a filtered data-set, according to certain example embodiments.

FIG. 6 is a flowchart illustrating a method for filtering data, according to certain example embodiments.

FIG. 7 is an illustration of a compressed bit-set, according to certain example embodiments.

FIG. 8 is an interface diagram of a graphical user interface to receive user inputs, according to certain example embodiments.

FIG. 9 is an illustration a presentation of a quantity, according to certain example embodiments.

FIG. 10 is a block diagram illustrating a representative software architecture, which may be used in conjunction

2

with various hardware architectures herein described and used to implement various embodiments.

FIG. 11 is a block diagram illustrating components of a machine, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

DETAILED DESCRIPTION

Graphical user interfaces (GUI) in are often configured to display presentations of filter criteria, such that a user of the GUI may provide inputs selecting one or more filter criteria directly into the GUI, and which a search or filtering system may then use to filter a data set (e.g., retrieve a set of search results). As the purpose of a GUI is ultimately to provide as much useful information to a user as possible, some similarly configured GUI may also provide an indication of a quantity associated with the filtered data set. For example, by presenting the user with an indication of a number of data objects among the filtered data set, the user might choose to add additional criteria to further filter the data set, or alternatively to remove filter criteria to expand the filtered data set. A number of technical hurdles may however limit the overall effectiveness of such features.

While the presentation of additional useful information is undeniably valuable, practical considerations for the execution of such features must be considered. For example, as a data set grows exponentially in size, the ability for a computing system to quickly and effectively determine an accurate count of a filtered data set may impose a significant burden on processors of the system, resulting in noticeable lag in display time. Systems to increase the efficiency and accuracy by which such information may be calculated and displayed would therefore be advantageous to users of such systems.

Thus, the present invention relates to improvements to systems and methods for filtering a data set. An audience filtering system is configured to perform operations that include accessing a database that comprises user profile data associated with a plurality of users, generating a bit-set based on the user profile data from the database, wherein the bit-set comprises an array of binary values, receiving a user input (e.g., via a GUI) that defines one or more filter criteria (e.g., a value of a user attribute), filtering the bit-set based on the one or more filter criteria, and identifying a subset of the plurality of users based on the filtered bit-set, according to certain example embodiments.

In some embodiments, the user profile data associated with the plurality of users comprises one or more user attributes. Such user attributes may for example include (but are not limited to): user demographics data (e.g., age, gender); location data; user activity data; device data; and language preferences. The audience filtering system may generate and cause display of a GUI that includes a presentation of one or more user selectable filtering criteria based on the user attributes from the database.

The audience filtering system generates one or more bit-sets based on the user attribute data. As discussed above, a bit-set, also known as a bit array, is an array data structure that compactly stores the user attribute data as a series of binary values, wherein each bit corresponds to a user attribute value among from the user attribute data. For example, the audience filtering system may generate the bit-set in response to receiving the user attribute data from each of the plurality of users, and may store the one or more bit-sets at a memory location within a database.

A user of the audience filtering system provides a user input that defines a set of attribute values of a set of user attributes. In some embodiments, the user may additionally provide Boolean operators to assign to the set of attribute values defined by the user input. In response to receiving the user input defining the one or more user attribute values, the audience filtering system retrieves one or more bit-sets associated with the set of user attributes, and filters the one or more bit-sets based on the set of attribute values. In instances where Boolean operators are also included, the audience filtering system may combine attribute values to filter the bit-sets based on the Boolean operators (e.g., and, or, etc.).

The audience filtering system identifies a subset of the plurality of users based on the filtered bit-sets. In some embodiments, the audience filtering system calculates a quantity of users among the subset of the plurality of users, and displays the quantity within the GUI in real-time. In this way, a user may provide a user input that defines a user attribute value, and the audience filtering system may calculate and update the quantity in real-time, thereby providing the user with an accurate indication of the quantity, more or less instantaneously. In some embodiments, the indication of the quantity may be displayed as a gauge, such as a linear gauge, or simply as a presentation of a numerical value.

In further embodiments, the audience filtering system receives a threshold value definition, wherein the threshold value definition defines a maximum or minimum quantity value. In such embodiments, in response to calculating the quantity of the subset of the plurality of users, the audience filtering system compares the quantity to the threshold value. In response to determining that the quantity transgresses the threshold value, the audience filtering system may generate and cause display of a notification alongside the presentation of the quantity within the GUI. For example, the notification may include a request to add or remove filter criteria.

Consider an illustrative example from a user perspective. The audience filtering system generates and causes display of a GUI at a client device, wherein the GUI comprises a presentation of a set of user selectable attribute values. For example, the GUI may include a display of a set of menus corresponding to each user attribute, wherein each menu comprises a display of the possible attribute values for any given attribute. A user of the audience filtering system selects attribute values via the GUI. In response to receiving the selections from the user, the audience filtering system filters a data set based on the attribute values selected, and causes display of an indication of a quantity of data objects within the filtered data set. Thus, the user is presented with an accurate count of data objects within the filtered data set, in real-time.

FIG. 1 is a block diagram showing an example messaging system 100 for exchanging data (e.g., messages and associated content) over a network. The messaging system 100 includes multiple client devices 102, each of which hosts a number of applications including a messaging client application 104. Each messaging client application 104 is communicatively coupled to other instances of the messaging client application 104 and a messaging server system 108 via a network 106 (e.g., the Internet).

Accordingly, each messaging client application 104 is able to communicate and exchange data with another messaging client application 104 and with the messaging server system 108 via the network 106. The data exchanged between messaging client applications 104, and between a messaging client application 104 and the messaging server

system 108, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system 108 provides server-side functionality via the network 106 to a particular messaging client application 104. While certain functions of the messaging system 100 are described herein as being performed by either a messaging client application 104 or by the messaging server system 108, it will be appreciated that the location of certain functionality either within the messaging client application 104 or the messaging server system 108 is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system 108, but to later migrate this technology and functionality to the messaging client application 104 where a client device 102 has a sufficient processing capacity.

The messaging server system 108 supports various services and operations that are provided to the messaging client application 104. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client application 104. In some embodiments, this data includes, message content, client device information, geolocation information, media annotation and overlays, message content persistence conditions, social network information, and live event information, as examples. In other embodiments, other data is used. Data exchanges within the messaging system 100 are invoked and controlled through functions available via GUIs of the messaging client application 104.

Turning now specifically to the messaging server system 108, an Application Program Interface (API) server 110 is coupled to, and provides a programmatic interface to, an application server 112. The application server 112 is communicatively coupled to a database server 118, which facilitates access to a database 120 in which is stored data associated with messages processed by the application server 112.

Dealing specifically with the Application Program Interface (API) server 110, this server receives and transmits message data (e.g., commands and message payloads) between the client device 102 and the application server 112. Specifically, the Application Program Interface (API) server 110 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client application 104 in order to invoke functionality of the application server 112. The Application Program interface (API) server 110 exposes various functions supported by the application server 112, including account registration, login functionality, the sending of messages, via the application server 112, from a particular messaging client application 104 to another messaging client application 104, the sending of media files (e.g., images or video) from a messaging client application 104 to the messaging server application 114, and for possible access by another messaging client application 104, the setting of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device 102, the retrieval of such collections, the retrieval of messages and content, the adding and deletion of friends to a social graph, the location of friends within a social graph, opening and application event (e.g., relating to the messaging client application 104).

The application server 112 hosts a number of applications and subsystems, including a messaging server application 114, an image processing system 116, a social network system 122, and an audience filtering system 124. The messaging server application 114 implements a number of

5

message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client application **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available, by the messaging server application **114**, to the messaging client application **104**. Other processor and memory intensive processing of data may also be performed server-side by the messaging server application **114**, in view of the hardware requirements for such processing.

The application server **112** also includes an image processing system **116** that is dedicated to performing various image processing operations, typically with respect to images or video received within the payload of a message at the messaging server application **114**.

The social network system **122** supports various social networking functions services, and makes these functions and services available to the messaging server application **114**. To this end, the social network system **122** maintains and accesses an entity graph **304** within the database **120**. Examples of functions and services supported by the social network system **122** include the identification of other users of the messaging system **100** with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.

The application server **112** is communicatively coupled to a database server **118**, which facilitates access to a database **120** in which is stored data associated with messages processed by the messaging server application **114**.

FIG. 2 is block diagram illustrating further details regarding the messaging system **100**, according to example embodiments. Specifically, the messaging system **100** is shown to comprise the messaging client application **104** and the application server **112**, which in turn embody a number of some subsystems, namely an ephemeral timer system **202**, a collection management system **204** and an annotation system **206**.

The ephemeral timer system **202** is responsible for enforcing the temporary access to content permitted by the messaging client application **104** and the messaging server application **114**. To this end, the ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, collection of messages (e.g., a SNAPCHAT story), or graphical element, selectively display and enable access to messages and associated content via the messaging client application **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

The collection management system **204** is responsible for managing collections of media (e.g., collections of text, image video and audio data). In some examples, a collection of content (e.g., messages, including images, video, text and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client application **104**.

The collection management system **204** furthermore includes a curation interface **208** that allows a collection

6

manager to manage and curate a particular collection of content. For example, the curation interface **208** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain embodiments, compensation may be paid to a user for inclusion of user generated content into a collection. In such cases, the curation interface **208** operates to automatically make payments to such users for the use of their content.

The annotation system **206** provides various functions that enable a user to annotate or otherwise modify or edit media content associated with a message. For example, the annotation system **206** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The annotation system **206** operatively supplies a media overlay (e.g., a SNAPCHAT filter) to the messaging client application **104** based on a geolocation of the client device **102**. In another example, the annotation system **206** operatively supplies a media overlay to the messaging client application **104** based on other information, such as, social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects, as well as animated facial models, such as those generated by the audience filtering system **124**. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay including text that can be overlaid on top of a photograph generated taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the annotation system **206** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **120** and accessed through the database server **118**.

In one example embodiment, the annotation system **206** provides a user-based publication platform that enables users to select a geolocation on a map, and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The annotation system **206** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In another example embodiment, the annotation system **206** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the annotation system **206** associates the media overlay of a highest bidding merchant with a corresponding geolocation for a predefined amount of time.

FIG. 3 is a block diagram illustrating components of the audience filtering system **124** that configure the audience filtering system **124** to generate a bit-set based on user attribute data, receive a user input defining one or more attribute values, filter the bit-set based on the one or more attribute values, identify a subset of a plurality of users

based on the filtered bit-set, and cause display of a count of a quantity of the subset of the plurality of users in real-time, according to certain example embodiments. The audience filtering system 124 is shown as including a bit-set module 302, a communication module 304, a filtering module 306, and presentation module 308, all configured to communicate with each other (e.g., via a bus, shared memory, or a switch). Any one or more of these modules may be implemented using one or more processors 310 (e.g., by configuring such one or more processors to perform functions described for that module) and hence include one or more of the processors 310.

Any one or more of the modules described may be implemented using hardware alone (e.g., one or more of the processors 310 of a machine) or a combination of hardware and software. For example, any module described of the audience filtering system 124 may physically include an arrangement of one or more of the processors 310 (e.g., a subset of or among the one or more processors of the machine) configured to perform the operations described herein for that module. As another example, any module of the audience filtering system 124 may include software, hardware, or both, that configure an arrangement of one or more processors 310 (e.g., among the one or more processors of the machine) to perform the operations described herein for that module. Accordingly, different modules of the audience filtering system 124 may include and configure different arrangements of such processors 310 or a single arrangement of such processors 310 at different points in time. Moreover, any two or more modules of the audience filtering system 124 may be combined into a single module, and the functions described herein for a single module may be subdivided among multiple modules. Furthermore, according to various example embodiments, modules described herein as being implemented within a single machine, database, or device may be distributed across multiple machines, databases, or devices.

FIG. 4 is a flowchart illustrating a method 400 for filtering data, according to certain example embodiments. Operations of the method 400 may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 4, the method 400 includes one or more operations 402, 404, 406, 408, and 410.

At operation 402, the bit-set module 302 accesses a database (e.g., the database 12) that comprises user profile data of a plurality of users. The user profile data associated with the plurality of users comprises one or more user attributes. Such user attributes may for example include (but are not limited to): user demographics data (e.g., age, gender); location data; user activity data; device data; and language preferences. At operation 404, the bit-set module 302 generates a bit-set based on the user attributes of the user profile data.

At operation 406, the communication module 304 receives a user input that defines one or more user attribute values of the user attributes from the database. For example, the audience filtering system 124 may generate and cause display of a GUI at a client device (e.g., client device 102), wherein the GUI includes a presentation of one or more user selectable filtering criteria based on the user attributes from the database 120. A user may provide a user input that includes selections of one or more user attribute values via the GUI.

In some embodiments, the user input that defines the user attribute values may also include one or more Boolean operators (e.g., AND, OR, NOT, AND NOT). The Boolean operators are used to connect the attribute values and define

filter criteria to filter the bit-sets. For example, by selecting an “AND,” the filtering module 306 filters data objects that include both of the attribute values. Similarly, by selecting “OR,” the filtering module 306 filters data objects that have one or the other (or both) of the attribute values).

At operation 408, the filtering module 306 filters the bit-set (of the corresponding user attribute values defined by the user input), based on the user attribute values, and at operation 410 identifies a subset of the plurality of users based on the filtered bit-set.

FIG. 5 is a flowchart illustrating a method 500 for causing display of an indication of a quantity of a filtered data-set, according to certain example embodiments. Operations of the method 500 may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 5, the method 500 includes one or more operations 502, 504, and 506 that may be performed as a part of (e.g., a subroutine of) one or more of the operations of the method 400, as seen in FIG. 4.

At operation 502, the filtering module 306 determines a quantity of users among the subset of the plurality of users identified based on the filtered bit-set. The quantity of users comprises a numerical value that indicates how many users (or data objects) are remaining within the filtered bit-set based on the filter criteria selected by the user.

At operation 504, the presentation module 308 determines that the quantity (the numerical value) of the subset of the plurality of users transgresses a predefined threshold value. For example, an administrator of the audience filtering system 124 may define one or more threshold values. The threshold values may define maximum or minimum values.

At operation 506, the presentation module 308 causes display of a presentation of the quantity at the client device 102. In some embodiments, the presentation may be based on whether or not the quantity transgresses the predefined threshold value, or which threshold value the quantity transgressed. For example, the threshold value may include a maximum value or a minimum value. In such embodiments, in response to determining that the quantity exceeds the threshold value, the presentation module 308 generates and causes display of the presentation of quantity of the quantity at the client device 102, wherein the presentation includes a notification based on the quantity transgressing the maximum value. For example, the notification may indicate that the user must add more filter criteria.

FIG. 6 is a flowchart illustrating a method 600 for filtering data, according to certain example embodiments. Operations of the method 600 may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 6, the method 600 includes one or more operations 602, 604, 606, and 608 that may be performed as a part of (e.g., a subroutine of) one or more of the operations of the method 400, as seen in FIG. 4.

At operation 602, the presentation module 308 causes display of a GUI that includes a presentation of a map image at the client device 102. In some embodiments, the presentation module 308 determines that the user attributes within the database 120 include location data that identify locations of users. In response to determining that the user attributes include location data, the presentation module 308 generates and causes display of a map image within the GUI at the client device 102.

At operation 604, the communication module 304 receives an identification of a geographic region. In some embodiments, the user input received at operation 406 of the method 400 may include an identification of the geographic region. For example, the user input may define a point (e.g.,

geographic coordinates, an address, a selection of a point within the map image), and a radius. In further embodiments, the user input may comprise a selection of the geographic region from within the map image, for example by drawing a shape (e.g., a circle) over the map image.

At operation 606, the filtering module 306 segments the geographic region into a set of geo-cells, wherein each geo-cell has a corresponding location identifier. For example, the filtering module 306 may determine a size of each geo-cell based on an overall size of the geographic region selected by the user input. In further embodiments, the geo-cells may each be of a predefined size and shape, and by selecting the geographic region the filtering module 306 simply selects any geo-cell located at least partially within a boundary of the geographic region. At operation 608, the filtering module 306 filters the bit-set based on the location identifiers of the geo-cells.

FIG. 7 is an illustration of a compressed bit-set 700, according to certain example embodiments. As seen in FIG. 7, each bit-set may correspond to a distinct attribute value of each user attribute from the user profile data within the database 120. Each bit-set therefore comprises a series of binary values, wherein each binary value indicates whether or not a particular user has the associated attribute value, and wherein a position of a binary value along the sequence identifies a particular user.

For example, as seen in FIG. 7 “User A” is depicted as having attribute values that include: female; aged 35+; and has an Apple device, whereas “User B” is depicted as having attribute values that include: male; aged between 13-17. A population of users may thereby be represented by a set of bit-sets.

In some embodiments, as discussed in operation 406 of the method 400, a user may provide a user input that defines one or more attribute values as well as one or more Boolean operators. The attribute values may thereby be combined to determine filter criteria for the bit-set based on the Boolean operators.

In some embodiments, the bit-set module 302 generates a data table based on the user profile data from the database 120. For example, as seen in FIG. 7, the bit-set module 302 may generate bit-sets for each user attribute value from among the user profile data, and store the bit-sets in a data table, wherein each row of the data table corresponds to a particular user attribute value, and each column of the data table corresponds to a user identifier.

FIG. 8 is an interface diagram of a graphical user interface 800 to receive user inputs, according to certain example embodiments. As seen in FIG. 8, the GUI 800 comprises a presentation of a set of user selectable user attribute values 802. A user of the audience filtering system 124 may provide a user input into the GUI (as discussed in operation 406 of the method 400), wherein the user input comprises a selection of one or more attribute values.

As seen in FIG. 8, the GUI 800 may also include a display 804 of a quantity of data objects within a filtered bit-set. As discussed in the method 500 of FIG. 5, the display 804 of the quantity is updated in real-time according to attribute values defined based on the user inputs.

FIG. 9 is an illustration a display 902 and 904 of the quantity of data objects within a filtered bit-set, according to certain example embodiments. As explained in the method 500 of FIG. 5, the displays of the quantity (902 and 904) may be updated to include a notification 906 and 908, wherein the notification is based on detecting a transgression of the quantity with a predefined threshold value.

Software Architecture

FIG. 10 is a block diagram illustrating an example software architecture 1006, which may be used in conjunction with various hardware architectures herein described. FIG. 10 is a non-limiting example of a software architecture and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture 1006 may execute on hardware such as machine 1100 of FIG. 11 that includes, among other things, processors 1104, memory 1114, and I/O components 1118. A representative hardware layer 1052 is illustrated and can represent, for example, the machine 1000 of FIG. 10. The representative hardware layer 1052 includes a processing unit 1054 having associated executable instructions 1004. Executable instructions 1004 represent the executable instructions of the software architecture 1006, including implementation of the methods, components and so forth described herein. The hardware layer 1052 also includes memory and/or storage modules memory/storage 1056, which also have executable instructions 1004. The hardware layer 1052 may also comprise other hardware 1058.

In the example architecture of FIG. 10, the software architecture 1006 may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture 1006 may include layers such as an operating system 1002, libraries 1020, applications 1016 and a presentation layer 1014. Operationally, the applications 1016 and/or other components within the layers may invoke application programming interface (API) API calls 1008 through the software stack and receive a response as in response to the API calls 1008. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/middleware 1018, while others may provide such a layer. Other software architectures may include additional or different layers.

The operating system 1002 may manage hardware resources and provide common services. The operating system 1002 may include, for example, a kernel 1022, services 1024 and drivers 1026. The kernel 1022 may act as an abstraction layer between the hardware and the other software layers. For example, the kernel 1022 may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services 1024 may provide other common services for the other software layers. The drivers 1026 are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers 1026 include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

The libraries 1020 provide a common infrastructure that is used by the applications 1016 and/or other components and/or layers. The libraries 1020 provide functionality that allows other software components to perform tasks in an easier fashion than to interface directly with the underlying operating system 1002 functionality (e.g., kernel 1022, services 1024 and/or drivers 1026). The libraries 1020 may include system libraries 1044 (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries 1020 may include API libraries 1046 such as media libraries (e.g., libraries to support presentation and manipulation of various media

11

format such as MPREG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **1020** may also include a wide variety of other libraries **1048** to provide many other APIs to the applications **1016** and other software components/modules.

The frameworks/middleware **1018** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **1016** and/or other software components/modules. For example, the frameworks/middleware **1018** may provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **1018** may provide a broad spectrum of other APIs that may be utilized by the applications **1016** and/or other software components/modules, some of which may be specific to a particular operating system **1002** or platform.

The applications **1016** include built-in applications **1038** and/or third-party applications **1040**. Examples of representative built-in applications **1038** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, and/or a game application. Third-party applications **1040** may include an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform, and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **1040** may invoke the API calls **1008** provided by the mobile operating system (such as operating system **1002**) to facilitate functionality described herein.

The applications **1016** may use built in operating system functions (e.g., kernel **1022**, services **1024** and/or drivers **1026**), libraries **1020**, and frameworks/middleware **1018** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems interactions with a user may occur through a presentation layer, such as presentation layer **1014**. In these systems, the application/component “logic” can be separated from the aspects of the application/component that interact with a user.

FIG. **11** is a block diagram illustrating components of a machine **1100**, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **11** shows a diagrammatic representation of the machine **1100** in the example form of a computer system, within which instructions **1110** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1100** to perform any one or more of the methodologies discussed herein may be executed. As such, the instructions **1110** may be used to implement modules or components described herein. The instructions **1110** transform the general, non-programmed machine **1100** into a particular machine **1100** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **1100** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as

12

a peer machine in a peer-to-peer (or distributed) network environment. The machine **1100** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1110**, sequentially or otherwise, that specify actions to be taken by machine **1100**. Further, while only a single machine **1100** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1110** to perform any one or more of the methodologies discussed herein.

The machine **1100** may include processors **1104**, memory **1106**, and I/O components **1118**, which may be configured to communicate with each other such as via a bus **1102**. The memory/storage **1106** may include a memory **1114**, such as a main memory, or other memory storage, and a storage unit **1116**, both accessible to the processors **1104** such as via the bus **1102**. The storage unit **1116** and memory **1114** store the instructions **1110** embodying any one or more of the methodologies or functions described herein. The instructions **1110** may also reside, completely or partially, within the memory **1114**, within the storage unit **1116**, within at least one of the processors **1104** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1100**. Accordingly, the memory **1114**, the storage unit **1116**, and the memory of processors **1104** are examples of machine-readable media.

The I/O components **1118** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1118** that are included in a particular machine **1100** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1118** may include many other components that are not shown in FIG. **11**. The I/O components **1118** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example embodiments, the I/O components **1118** may include output components **1126** and input components **1128**. The output components **1126** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1128** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

13

In further example embodiments, the I/O components **1118** may include biometric components **1130**, motion components **1134**, environmental environment components **1136**, or position components **1138** among a wide array of other components. For example, the biometric components **1130** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1134** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environment components **1136** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1138** may include location sensor components (e.g., a Global Position system (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1118** may include communication components **1140** operable to couple the machine **1100** to a network **1132** or devices **1120** via coupling **1122** and coupling **1124** respectively. For example, the communication components **1140** may include a network interface component or other suitable device to interface with the network **1132**. In further examples, communication components **1140** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1120** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

Moreover, the communication components **1140** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1140** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1140**, such as, location via Internet Protocol (IP) geo-location, location via Wi-Fi®

14

signal triangulation, location via detecting a NFC beacon signal that may indicate a particular location, and so forth.

Glossary

“CARRIER SIGNAL” in this context refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible medium to facilitate communication of such instructions. Instructions may be transmitted or received over the network using a transmission medium via a network interface device and using any one of a number of well-known transfer protocols.

“CLIENT DEVICE” in this context refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“COMMUNICATIONS NETWORK” in this context refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (ENDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard setting organizations, other long range protocols, or other data transfer technology.

“EMPIERICAL MESSAGE” in this context refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

“MACHINE-READABLE MEDIUM” in this context refers to a component, device or other tangible media able to store instructions and data temporarily or permanently and may include, but is not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache

memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., code) for execution by a machine, such that the instructions, when executed by one or more processors of the machine, cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

“COMPONENT” in this context refers to a device, physical entity or logic having boundaries defined by function or subroutine calls, branch points, application program interfaces (APIs), or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware components are

temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an Application Program Interface (API)). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components may be distributed across a number of geographic locations.

“PROCESSOR” in this context refers to any circuit or virtual circuit (a physical circuit emulated by logic executing

17

on an actual processor) that manipulates data values according to control signals (e.g., “commands”, “op codes”, “machine code”, etc.) and which produces corresponding output signals that are applied to operate a machine. A processor may, for example, be a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC) or any combination thereof. A processor may further be a multi-core processor having two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously.

“TIMESTAMP” in this context refers to a sequence of characters or encoded information identifying when a certain event occurred, for example giving date and time of day, sometimes accurate to a small fraction of a second.

“LIFT” in this context is a measure of the performance of a targeted model at predicting or classifying cases as having an enhanced response (with respect to a population as a whole), measured against a random choice targeting model.

“PHONEME ALIGNMENT” in this context, a phoneme is a unit of speech that differentiates one word from another. One phoneme may consist of a sequence of closure, burst, and aspiration events; or, a diphthong may transition from a back vowel to a front vowel. A speech signal may therefore be described not only by what phonemes it contains, but also the locations of the phonemes. Phoneme alignment may therefore be described as a “time-alignment” of phonemes in a waveform, in order to determine an appropriate sequence and location of each phoneme in a speech signal.

“AUDIO-TO-VISUAL CONVERSION” in this context refers to the conversion of audible speech signals into visible speech, wherein the visible speech may include a mouth shape representative of the audible speech signal.

“TIME DELAYED NEURAL NETWORK (TDNN)” in this context, a TDNN is an artificial neural network architecture whose primary purpose is to work on sequential data. An example would be converting continuous audio into a stream of classified phoneme labels for speech recognition.

“BI-DIRECTIONAL LONG-SHORT TERM MEMORY (BLSTM)” in this context refers to a recurrent neural network (RNN) architecture that remembers values over arbitrary intervals. Stored values are not modified as learning proceeds. RNNs allow forward and backward connections between neurons. BLSTM are well-suited for the classification, processing, and prediction of time series, given time lags of unknown size and duration between events.

What is claimed is:

1. A method comprising:

accessing a database comprising user profile data of a plurality of users, the user profile data including user attribute data;
determining one or more user attributes based on the user attribute data of the user profile data;
generating a graphical user interface that comprises a display of one or more menu elements that correspond with each of the one or more user attributes, the one or more menu elements comprising one or more selectable filter criteria selected from among a plurality of filter criteria based on the one or more user attributes;
receiving a selection of one or more filter criteria from among the display of the one or more selectable filter criteria, the selection including a Boolean operator;

18

identifying a subset of the plurality of users based on a user attribute associated with the one or more filter criteria;

generating a bit-set based on the user attribute data of the user profile data;

receiving a user input, the user input that defines a minimum threshold value for a quantity of users in the subset;

determining that a quantity of the subset of the plurality of users is below the threshold value received by the user input based on a comparison of a count of user identifiers in the filtered bit-set against the minimum threshold value stored in the data base; and

responsive to determining that the quantity of the subset of the plurality of users is below the threshold value, causing display of a notification that includes a presentation of the quantity of the subset of the plurality of users, and a display of a message selected from among a plurality of messages based on the determining that the quantity is below the threshold value, the presentation of the quantity of the subset of the plurality of users comprising a linear gauge that indicates the quantity of the subset of the plurality of users.

2. The method of claim 1, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects a geographic region; and
identifying the subset of the plurality of users based on the user attribute and the geographic region.

3. The method of claim 1, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects the user attribute.

4. The method of claim 1, wherein the user attribute includes a user demographic.

5. The method of claim 1, wherein the user attribute includes a device attribute.

6. A system comprising:

a memory; and

at least one hardware processor coupled to the memory and comprising instructions that causes the system to perform operations comprising:

accessing a database comprising user profile data of a plurality of users, the user profile data including user attribute data;

determining one or more user attributes based on the user attribute data of the user profile data;

generating a graphical user interface that comprises a display of one or more menu elements that correspond with each of the one or more user attributes, the one or more menu elements comprising one or more selectable filter criteria selected from among a plurality of filter criteria based on the one or more user attributes;

receiving a selection of one or more filter criteria from among the display of the one or more selectable filter criteria, the selection including a Boolean operator;

identifying a subset of the plurality of users based on a user attribute associated with the one or more filter criteria;

generating a bit-set based on the user attribute data of the user profile data;

receiving a user input, the user input that defines a minimum threshold value for a quantity of users in the subset;

determining that a quantity of the subset of the plurality of users is below the threshold value received by the user input based on a comparison of a count of user identifiers in the filtered bit-set against the minimum threshold value stored in the data base; and

19

responsive to determining that the quantity of the subset of the plurality of users is below the threshold value, causing display of a notification that includes a presentation of the quantity of the subset of the plurality of users, and a display of a message selected from among a plurality of messages based on the determining that the quantity is below the threshold value, the presentation of the quantity of the subset of the plurality of users comprising a linear gauge that indicates the quantity of the subset of the plurality of users.

7. The system of claim 6, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects a geographic region; and identifying the subset of the plurality of users based on the user attribute and the geographic region.

8. The system of claim 6, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects the user attribute.

9. The system of claim 6, wherein the user attribute includes a user demographic.

10. The system of claim 6, wherein the user attribute includes a device attribute.

11. A non-transitory machine-readable storage medium comprising instructions that, when executed by one or more processors of a machine, cause the machine to perform operations comprising:

accessing a database comprising user profile data of a plurality of users, the user profile data including user attribute data;

determining one or more user attributes based on the user attribute data of the user profile data;

generating a graphical user interface that comprises a display of one or more menu elements that correspond with each of the one or more user attributes, the one or more menu elements comprising one or more selectable filter criteria selected from among a plurality of filter criteria based on the one or more user attributes;

receiving a selection of one or more filter criteria from among the display of the one or more selectable filter criteria, the selection including a Boolean operator;

20

identifying a subset of the plurality of users based on a user attribute associated with the one or more filter criteria;

generating a bit-set based on the user attribute data of the user profile data;

receiving a user input, the user input that defines a minimum threshold value for a quantity of users in the subset;

determining that a quantity of the subset of the plurality of users is below the threshold value received by the user input based on a comparison of a count of user identifiers in the filtered bit-set against the minimum threshold value stored in the data base; and

responsive to determining that the quantity of the subset of the plurality of users is below the threshold value, causing display of a notification that includes a presentation of the quantity of the subset of the plurality of users, and a display of a message selected from among a plurality of messages based on the determining that the quantity is below the threshold value, the presentation of the quantity of the subset of the plurality of users comprising a linear gauge that indicates the quantity of the subset of the plurality of users.

12. The non-transitory machine-readable storage medium of claim 11, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects a geographic region; and identifying the subset of the plurality of users based on the user attribute and the geographic region.

13. The non-transitory machine-readable storage medium of claim 11, wherein the identifying the subset of the plurality of users further comprises:

receiving an input that selects the user attribute.

14. The non-transitory machine-readable storage medium of claim 11, wherein the user attribute includes a user demographic.

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